SolarJEPA

Solar X-ray time-series analysis

GOES-16 XRS-A / XRS-B
May–August 2024

Objective: characterize solar observations and demonstrate reproducible 30-minute classical forecasting

Data collection

GOES-16 EXIS/XRS science products at one-minute cadence

123

pinned daily files

177,120

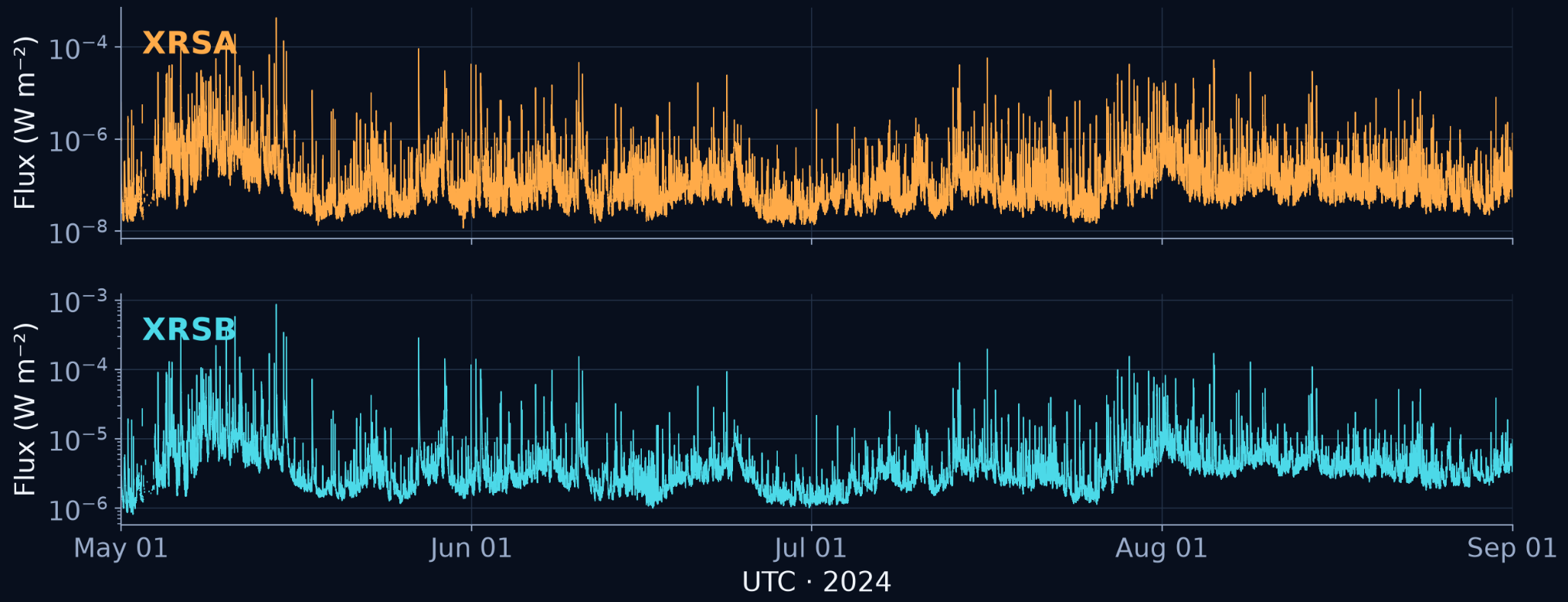
recorded minute bins

Month (2024)	May	June	July	August
Daily files	31	30	31	31
Minute bins	44,640	43,200	44,640	44,640

UTC bin starts / Irradiance in W/m² / Four manifest-pinned collections

Solar X-ray flux over time

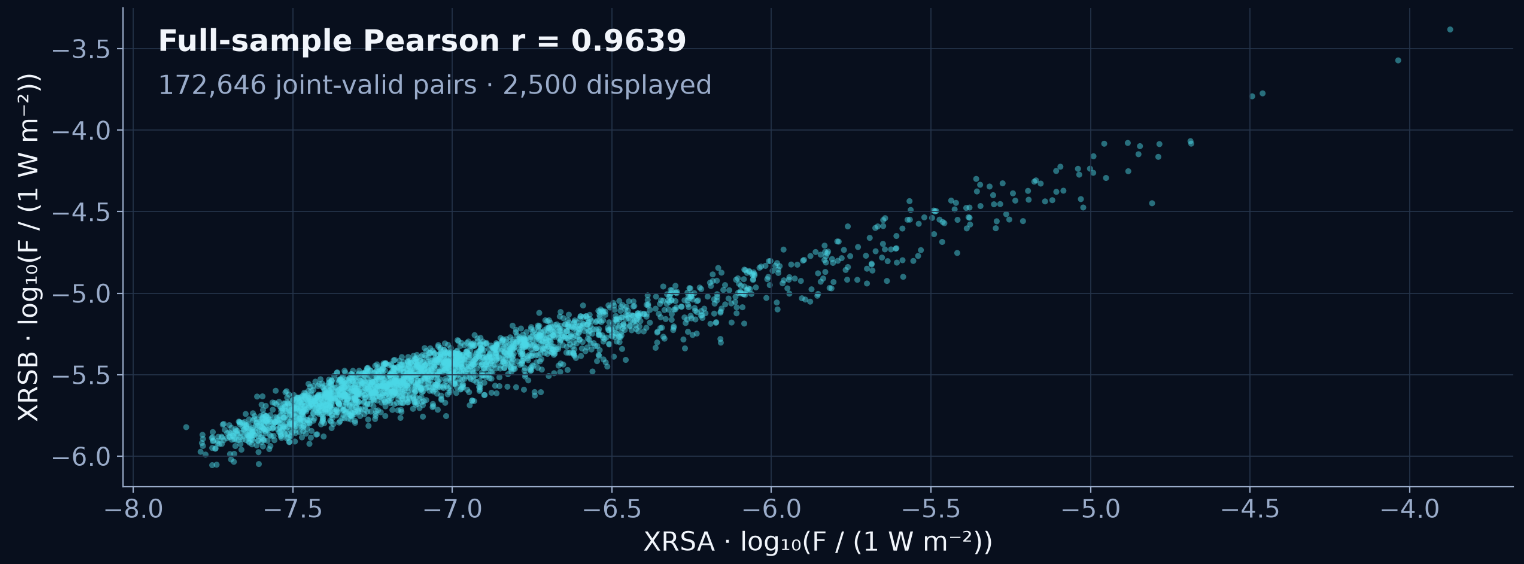
Full May–August archive with QC-rejected values retained as gaps



Transient peaks and a changing background motivate local diagnostics

Paired channel behavior

Log irradiance scatter from jointly valid one-minute observations



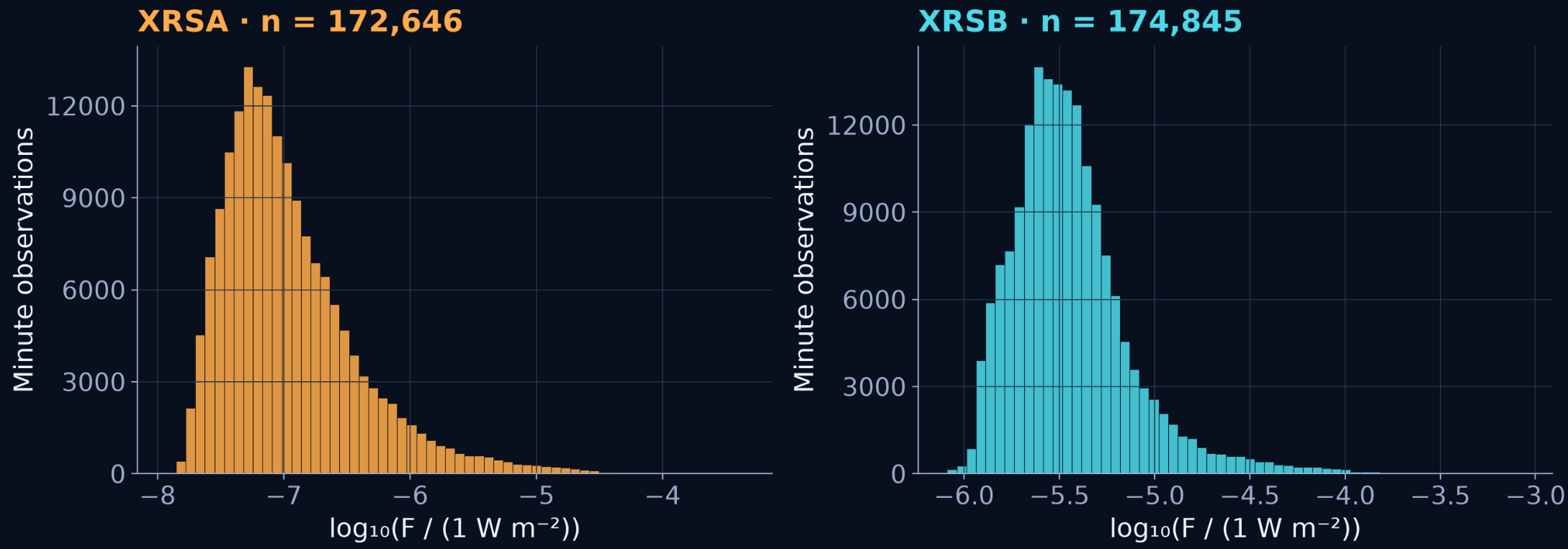
0.9639

Pearson correlation
172,646 paired minutes

2,500 pairs displayed
Full-sample statistic

Flux distributions

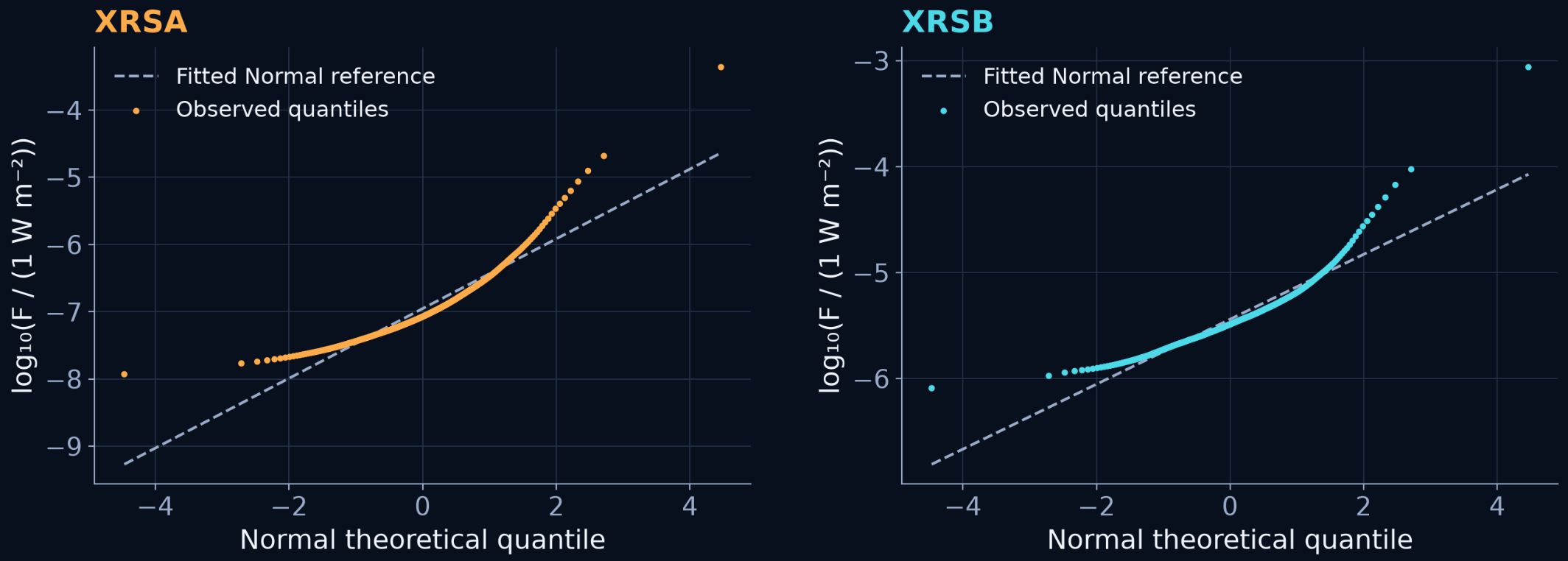
Histograms use each channel’s full QC-valid May–August population



Log scaling makes the dynamic range easier to inspect

Normal Q–Q diagnostics

Marginal log distributions show departures from a fitted normal reference



Tail departures caution against a simple Gaussian marginal assumption

Descriptive statistics and missing values

Unlogged QC-valid irradiance in W/m², May–August 2024

Statistic	XRS-A	XRS-B
Valid bins	172,646	174,845
Mean	5.245e-7	5.663e-6
Median	8.462e-8	3.252e-6
Sample std.	4.178e-6	1.416e-5
Minimum	1.177e-8	8.122e-7
Maximum	4.365e-4	8.688e-4

0 absent timestamps

QC-rejected values
A: 4,474 (2.53%)
B: 2,275 (1.28%)

No imputation
NaN gaps and masks persist

Complete timestamp coverage still includes unusable measurements

Stationarity: ADF and KPSS

10,149 consecutive June training minutes, with 5% decision threshold

Series	ADF statistic	ADF p	KPSS statistic	KPSS p
A: log level	−8.763	2.64×10^{-14}	0.554	0.02944
A: Δlog	−22.395	≈ 0*	0.00245	> 0.10
B: log level	−8.224	6.32×10^{-13}	0.658	0.01734
B: Δlog	−21.191	≈ 0*	0.00273	> 0.10

ADF null: unit root

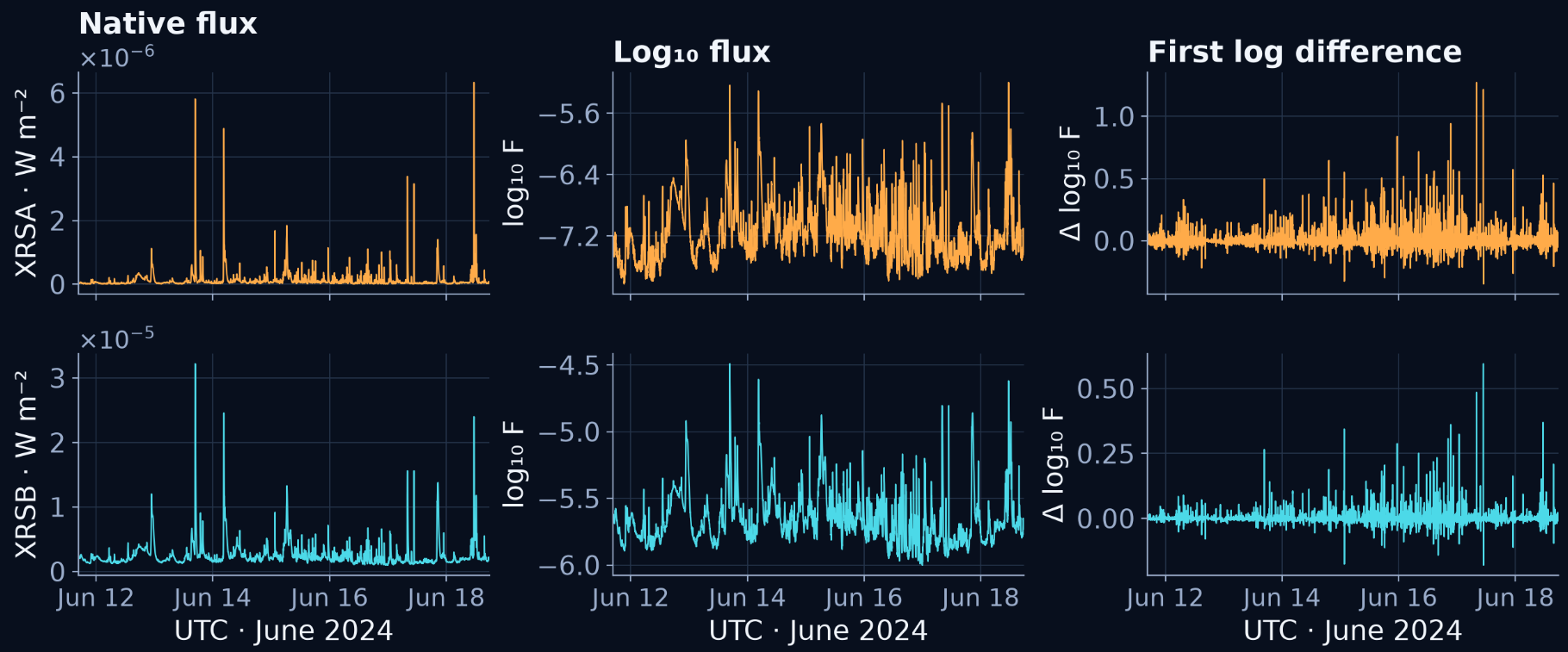
KPSS null: level stationarity

First differences support the local d = 1 demonstration

* Numerical p-value cutoff. Non-rejection does not prove global stationarity.

Log transformation and differencing

Before and after views use the same complete June training segment

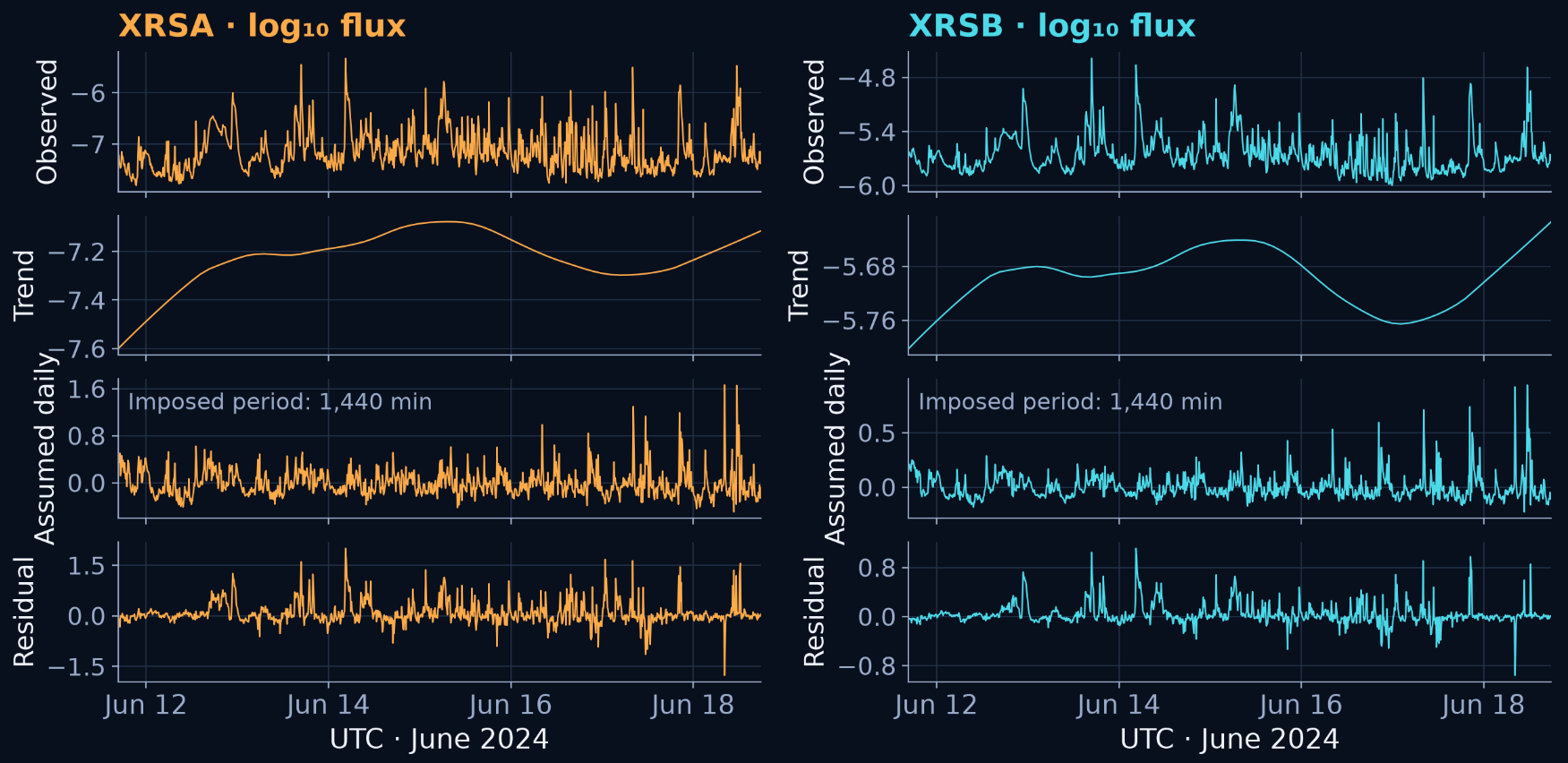


$y_t = \log_{10}(F_t / (1\ \mathrm{W\ m^{-2}}))$

$\Delta y_t = y_t - y_{t-1}$

Exploratory STL decomposition

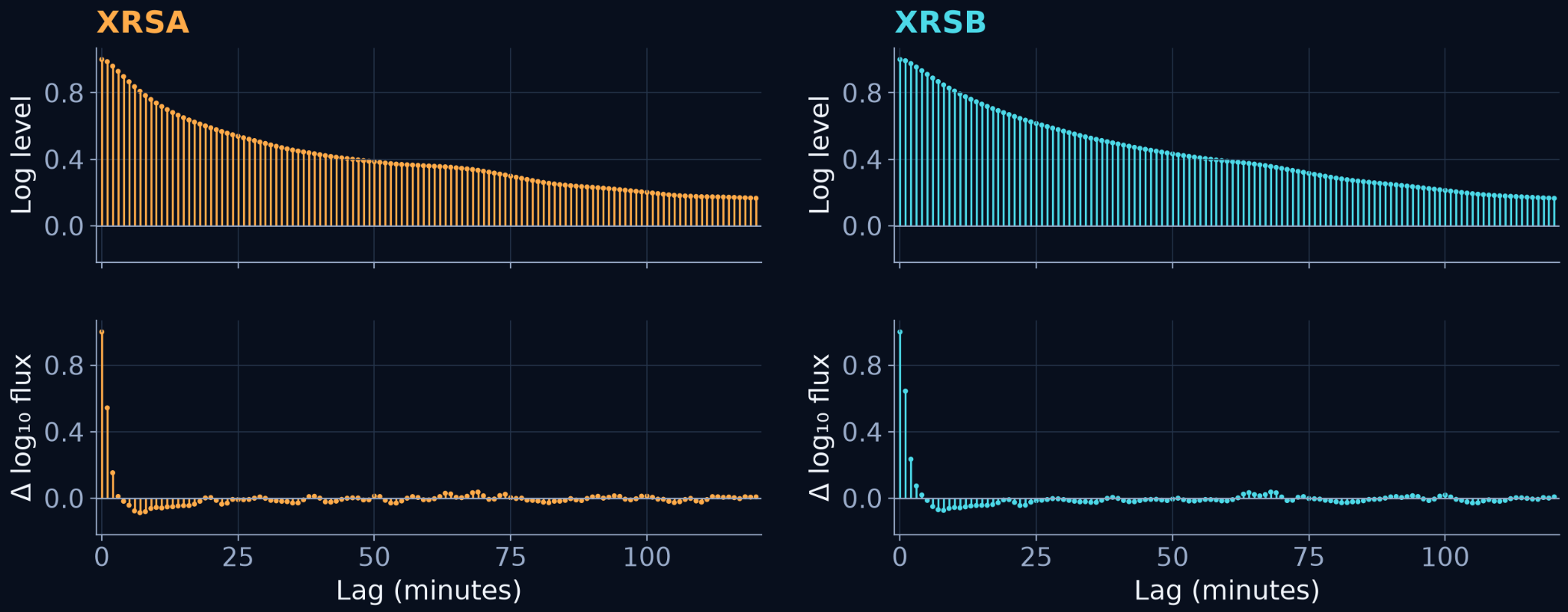
Robust STL with an assumed period of 1,440 minutes



An imposed daily component does not establish stable solar seasonality

Autocorrelation

ACF of log levels and first log differences, with lag units in minutes



Differencing reduces level persistence but leaves short-lag structure

Partial autocorrelation and lag selection

PACF guides a bounded candidate set. Training BIC makes the order choice



XRS-A: (p, d, q) = (1, 1, 0)

XRS-B: (p, d, q) = (2, 1, 0)

Classical model identification

Seven fixed candidates per channel on the same 2,880 June training minutes

Family	Candidate specification	Role
AR	AR(1), AR(2)	Past observations
MA	MA(1), MA(2)	Past innovations
ARMA	ARMA(1,1)	Both components
ARIMA	(1,1,0), (2,1,0)	First log difference
SARIMA	Deferred	Stable period unestablished

All 14 nonseasonal fits converged. No seasonal fit or score is claimed

AIC/BIC comparison and selection

Comparable d = 1 candidates only, ranked separately for each channel

Channel / order	AIC	BIC	ΔBIC within channel
A / (1,1,0)	**-12311.066**	**-12299.136**	**0.000**
A / (2,1,0)	-12310.261	-12292.365	6.771
B / (1,1,0)	-20616.476	-20604.546	64.286
B / (2,1,0)	**-20686.728**	**-20668.832**	**0.000**

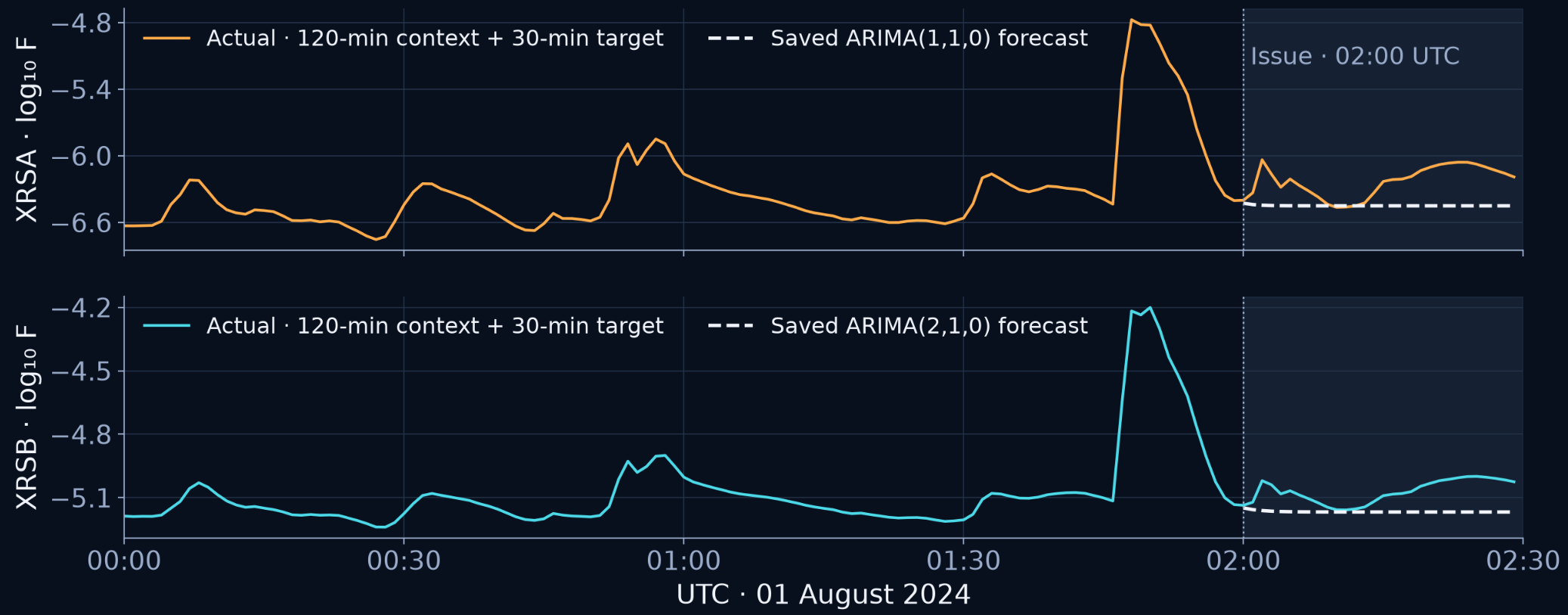
$$\mathrm{AIC} = -2\ell + 2k$$

$$\mathrm{BIC} = -2\ell + k \ln(n_{\mathrm{eff}})$$

Selected: A (1,1,0) and B (2,1,0) by training BIC

Actual versus predicted irradiance

1 August 2024, 02:00 UTC. Axes show $\log_{10}(F / (1\ \mathrm{W\ m^{-2}}))$ in dex



120 observed past minutes and a 30-minute forecast horizon

Forecast demonstration results

One retrospective held-out 30-minute example

Channel	Selected order	Log MAE (dex)	Log RMSE (dex)
XRS-A	(1,1,0)	0.218835	0.256635
XRS-B	(2,1,0)	0.094347	0.109055

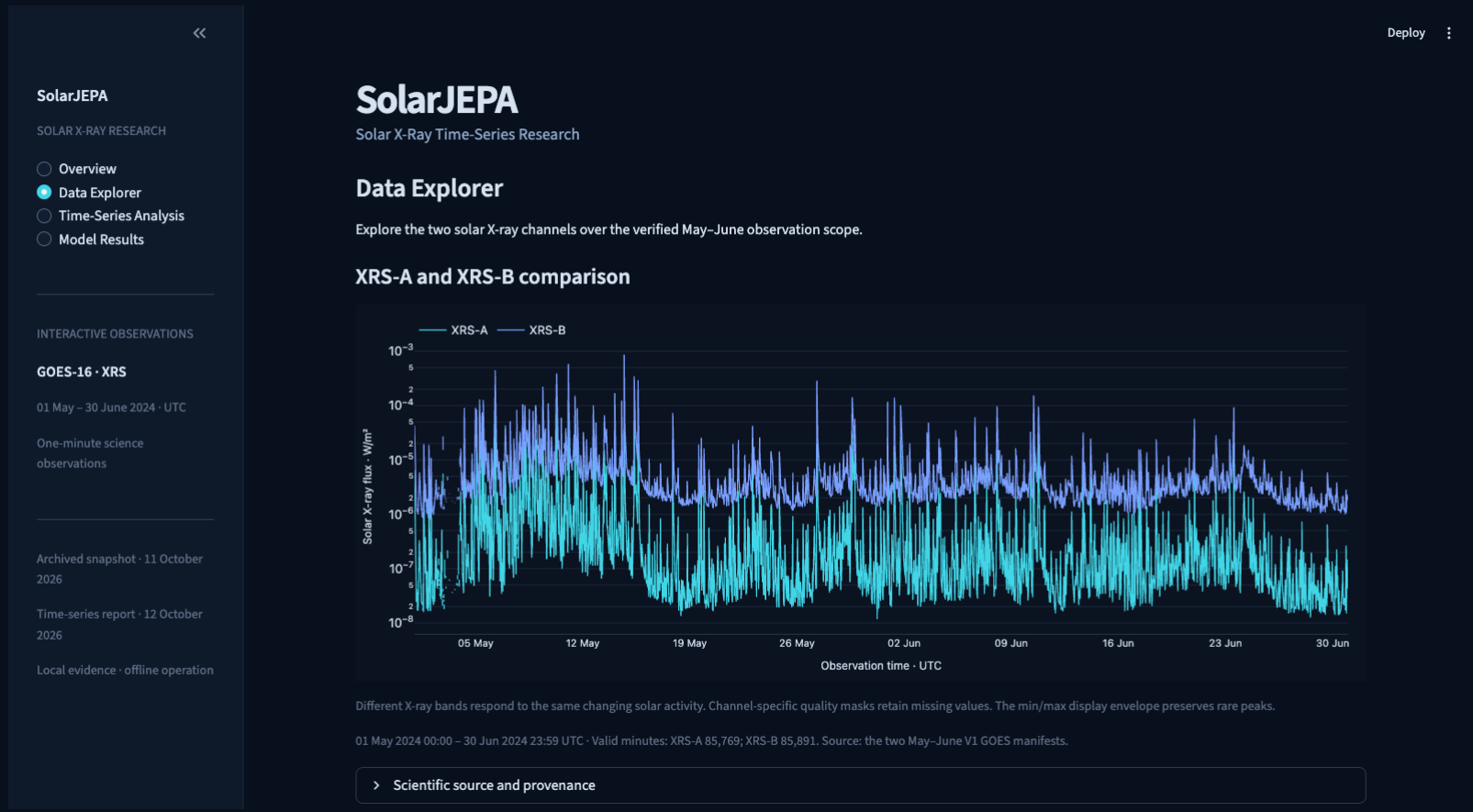
$$\mathrm{MAE} = (1/n)\,\Sigma\,|e_t| \qquad \mathrm{RMSE} = \sqrt{[(1/n)\,\Sigma\,e_t^2]}$$

n = 30 target minutes per channel. Errors use $\log_{10}$ irradiance.

These values do not represent overall forecasting skill

Local Streamlit deployment

Thin presentation layer over reusable scientific and inference modules



Scientific core

Pinned data and QC
Reusable diagnostics

Saved inference

Frozen parameters
Past context only

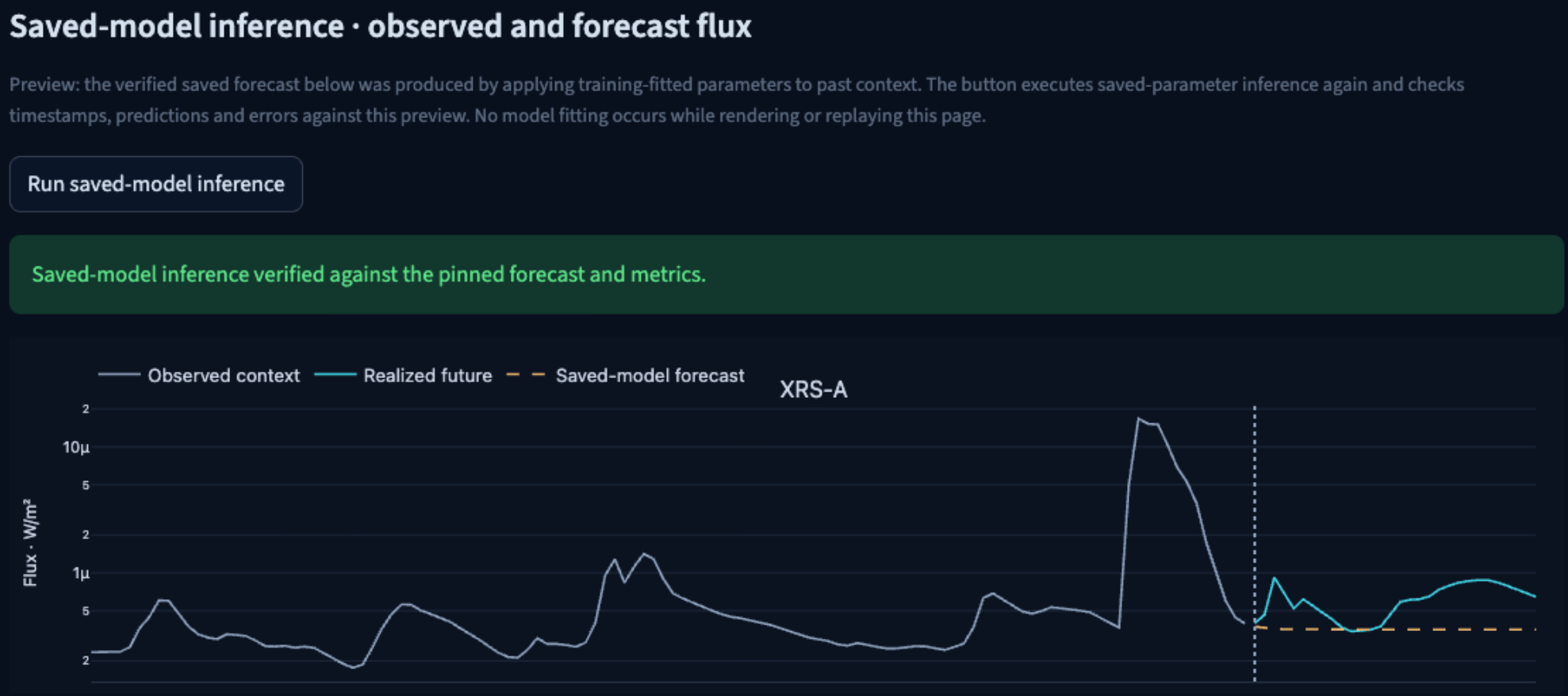
Streamlit views

Charts and tables

Screenshot: original interactive observation scope is May–June 2024

Saved-model inference in the dashboard

Detail of the actual Streamlit replay action and XRS-A forecast



Issue: 1 August 2024, 02:00 UTC. 120 past bins, 30 target bins

Conclusion and future improvements

A reproducible classical analysis with explicit scientific boundaries

What the evidence supports

Verified GOES data and diagnostics
Bounded classical model comparison
Reproducible local inference

What needs further evaluation

Many chronological forecast origins
Residual checks and uncertainty
Regime robustness and delivery latency

SARIMA awaits a defensible stable period

Operational solar-flare forecasting remains unestablished

Professor’s 11-requirement summary

All topics covered, with the SARIMA applicability decision disclosed

Requirement	Slides	Coverage
1 Data collection	2	Verified
2 EDA and descriptive statistics	3–7	Actual plots and tables
3 Missing values	7	QC masks, no imputation
4 Stationarity testing	8	ADF + KPSS
5 Transformations	9	Log + first difference
6 Seasonal decomposition	10	Exploratory STL
7 ACF / PACF	11–12	Actual dependence plots
8 Model identification	13–14	Nonseasonal fits, SARIMA deferred
9 Lag selection	12–14	p, d, q and training BIC
10 AIC / BIC	14	Compatible comparisons
11 Model deployment	17–18	Local saved-model inference

Ten requirements complete. #8 remains partial under an all-family fitting rubric.